

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 3 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTCVC305 - SURVEYING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which of the following is NOT a fundamental principle of chain surveying?
 - A) Working from whole to part
 - B) Fixing a point from at least two measurements
 - C) Always using a steel tape for maximum accuracy
 - D) Establishing a network of well-conditioned triangles
 - 2 A triangle in chain surveying is considered well-conditioned when its angles are approximately
 - A) 30° , 60° , 90°
 - B) All angles $< 30^\circ$
 - C) None of the angles is less than 30° or more than 120°
 - D) All angles are equal
 - 3 Ranging is the process of
 - A) Measuring angles
 - B) Establishing intermediate points on a straight line
 - C) Setting perpendiculars
 - D) Plotting the survey
 - 4 In the Whole Circle Bearing (WCB) system, the bearing of a line is measured:
 - A) Clockwise or anti-clockwise from the North or South meridian
 - B) Clockwise from the North meridian
 - C) Anti-clockwise from the North meridian
 - D) Clockwise from the East meridian
 - 5 The prismatic compass gives bearings in
 - A) Whole circle bearing system
 - B) Quadrantal system
 - C) Both A and B
 - D) Reduced bearing only

- 6 Which method of plane table surveying is best suited for establishing the position of inaccessible points?
 - A) Radiation
 - B) Intersection
 - C) Resection
 - D) Traversing
- 7 A level line is a line that is
 - A) Parallel to the ground
 - B) Curved and parallel to the mean spheroidal surface
 - C) Straight and horizontal
 - D) Straight and inclined
- 8 Which of the following describes a 'contour line'?
 - A) A line connecting points of equal elevation
 - B) A line connecting points of equal horizontal distance
 - C) A line representing a property boundary
 - D) A line indicating the direction of water flow
- 9 The process of bringing the vertical axis of the theodolite truly vertical is called:
 - A) Centering
 - B) Leveling
 - C) Focusing
 - D) Traversing
- 10 Vertical angle is the angle between
 - A) True meridian and survey line
 - B) Horizontal plane and line of sight
 - C) Magnetic meridian and line of sight
 - D) Two horizontal lines
- 11 Reconnaissance survey is conducted to
 - A) Fix detailed alignment
 - B) Collect general information about an area
 - C) Set out final works
 - D) Estimate final quantities
- 12 The primary purpose of a 'route survey' for highways is to:
 - A) Determine the traffic volume
 - B) Locate the most economical and practical alignment for the road
 - C) Assess the environmental impact
 - D) Design the pavement structure

Q2. Solve the following.

- A) Explain the fundamental principles of chain surveying and discuss the importance of "working from whole to part".
- B) The area of the plan of an old survey plotted to a scale of 10 metres to 1 cm measures now as 100.2 sq. cm as found by a planimeter. The plan is found to have shrunk so that a line originally 10 cm long now measures 9.7 cm only. Find (i) the shrunk scale, (ii) true area of the survey.

Q3. Solve the following.

- A) Compare and contrast the Prismatic Compass and the Surveyor's Compass, highlighting their key differences in reading mechanisms and bearing systems.

- B) The following bearings were taken in running a closed compass traverse while 6 surveying in Jhansi, Allahabad:

Line	F.B.	B.B.
AB	48°25'	230°00'
BC	177°45'	356°00'
CD	104°15'	284°55'
DE	165°15'	345°15'
EA	259°30'	79°00'

- (i) State the stations which are affected by local attraction and by how much.
(ii) Determine the correct bearings.
(iii) Calculate the true bearings, if the declination was 1°30' W.

Q4. Solve Any Two of the following.

- A) Define 'level line,' 'horizontal line,' and 'datum' in the context of levelling with 6 proper diagram.
B) Following is the page of a level field book. The readings in the level book were 6 written with pencil and some of these got erased. The erased readings are marked with question marks. Calculate the missing readings.

Station	B.S.	I.S.	F.S.	Rise	Fall	R.L.	Remarks
1	?					150.00	B.M.
2		2.457			0.827	?	
3		2.400		0.057		?	
4	2.697		?		?	148.070	C.P.
5	?		2.051	0.646		148.716	
6		2.500		1.068		149.784	C.P.
7		2.896			?	149.388	
8		?			0.124	?	
9			2.672	0.348		149.612	

- C) Define 'contour' and explain the main 'characteristics of contours.' How are 6 contour maps useful in engineering and planning applications?

Q5. Solve Any Two of the following.

- A) Describe the main components of a vernier transit theodolite. Explain the 6 "temporary adjustments" that must be performed before taking any observation with the instrument.
B) Explain the concept of "traversing" using a theodolite. Discuss the terms "closing 6 error" and "distribution of closing error" in the context of a closed traverse.
C) Explain the difference between "vernier theodolite" and "micro-optic theodolite." 6

Q6. Solve Any Two of the following.

- A) Explain: i) Reconnaissance, ii) Preliminary Survey, and iii) Location Survey. 6
B) What is "setting out works" in engineering surveying? 6

- C) What is tunnel correlation? Explain the methods used for connecting underground and surface surveys. 6

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